DAY 2 OF 30

# OneLake: one copy of your data

The storage layer under everything.
And why it changes how teams share data.

Shortcuts | Delta | Workspaces

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THE PROBLEM

# Every team kept its own copy of the data.

| | | |
|---|---|---|
| 1 | **Finance** | Its own storage account |
| 2 | **Marketing** | Another lake, another copy |
| 3 | **Data science** | Exports on a shared drive |
| 4 | **Reporting** | One more copy in the BI tool |

Four copies, four versions of the truth. **Nobody knows which one is current.**

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THE SIMPLE VERSION

# OneLake is OneDrive for data.

Your company has one OneDrive, not one per team. Everyone works in it, and permissions decide who sees what.

OneLake is the same idea for analytics data. **One lake per organisation**, already there on day one.

| | |
|---|---|
| ✓ **One per tenant** | Nothing to create or size |
| ✓ **Familiar layout** | Workspaces, then items |
| ✓ **One format** | Tables are stored as Delta |

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HOW IT WORKS

# How OneLake is organised

1. **Tenant**: the whole organisation, one lake
2. **Workspace**: a folder for a team or project
3. **Item**: a lakehouse, warehouse or database
4. **Inside a lakehouse**: Tables and Files

**Shortcuts**
A pointer to data that lives somewhere else. It appears in the lake without being copied.

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FOR ENGINEERS

# What matters under the hood

### It speaks ADLS Gen2

OneLake supports the ADLS Gen2 APIs and SDKs, so tools built for Data Lake Storage keep working.

### Region follows capacity

A workspace's data lives in the region of the capacity it is assigned to.

### Storage is billed on its own

Stored gigabytes are charged separately from the compute capacity.

### Open table formats

Fabric engines write Delta Parquet. OneLake can also present tables as Iceberg for other engines.

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THE HONEST PART

# Does one lake mean no silos?

| IT HELPS WHEN | IT WILL NOT FIX |
| --- | --- |
| Teams need the same data without copying it | Unclear ownership of each dataset |
| Data is spread across Azure, AWS and Google | A messy workspace structure |
| You want one place to govern access | Poor data quality at the source |

> **The catch:** one lake can still become one swamp. Decide workspace boundaries and who owns each table before teams start loading data.

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FROM REAL WORK

# OneLake on a real project

A people-analytics platform where employee data was split across many files.

1. **Performance, ratings and learning data** landed in one Lakehouse
2. **Semantic models** read from that single copy
3. **Role-based access and row-level security** protected it
4. **Daily pipelines** kept it fresh

The value was not the storage. It was **one agreed copy**.

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SAVE THIS

# OneLake in five lines

1. **One lake** per organisation, created for you
2. **Workspaces** are the folders, items live inside
3. **Tables are Delta**, so every engine reads them
4. **Shortcuts** bring in outside data with no copy
5. **Ownership** still has to be decided by people

## Tomorrow: Workspaces and capacities. How Fabric is organised, and how you pay for it.

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Follow for Day 3